

Chopsticks & Kimchi Project

Steven B., Laura G., Lucas L., Tim W.



**CHOPSTICKS
& KIMCHI**
PROJECT

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Timeline

Mechanical

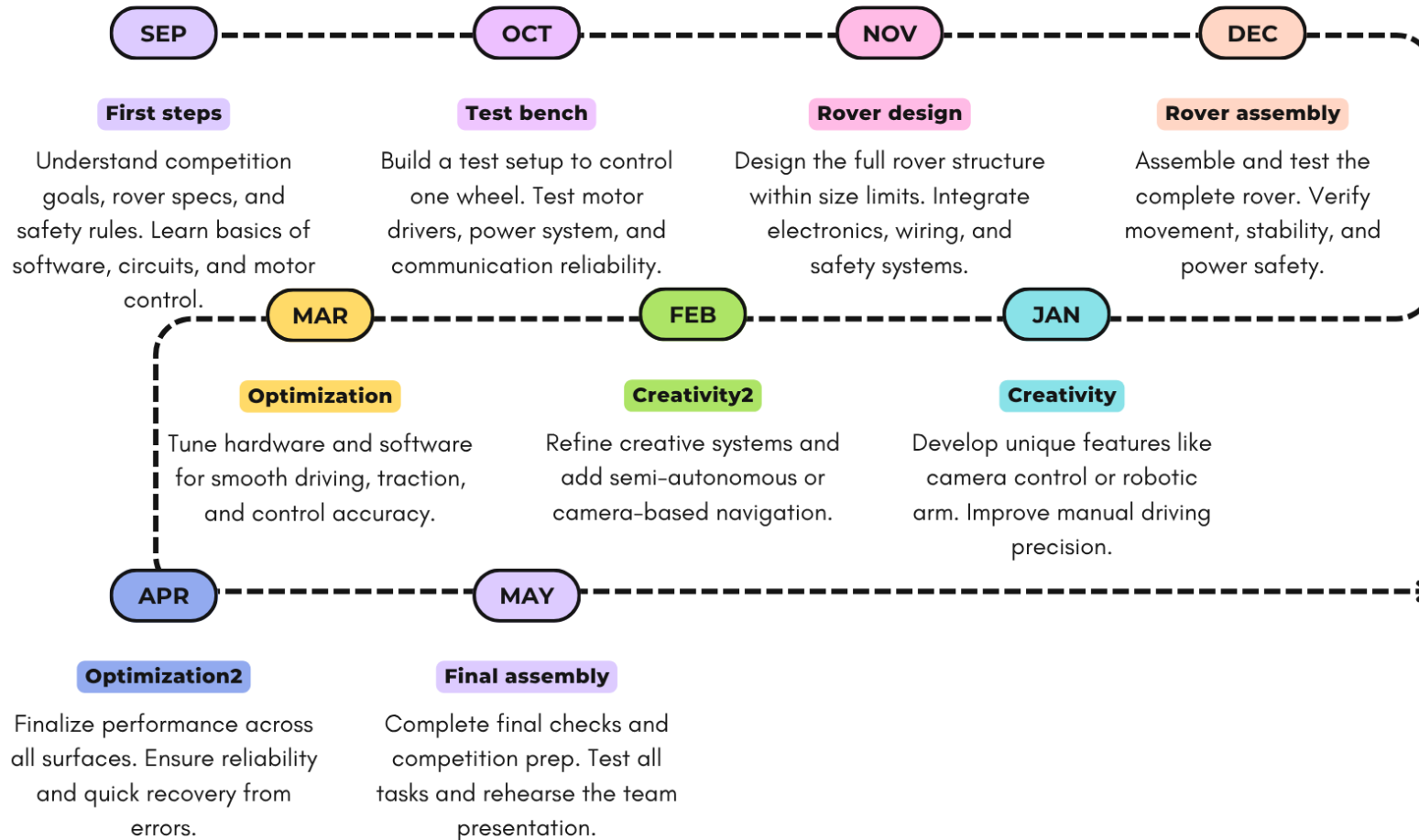
Electrical

Software

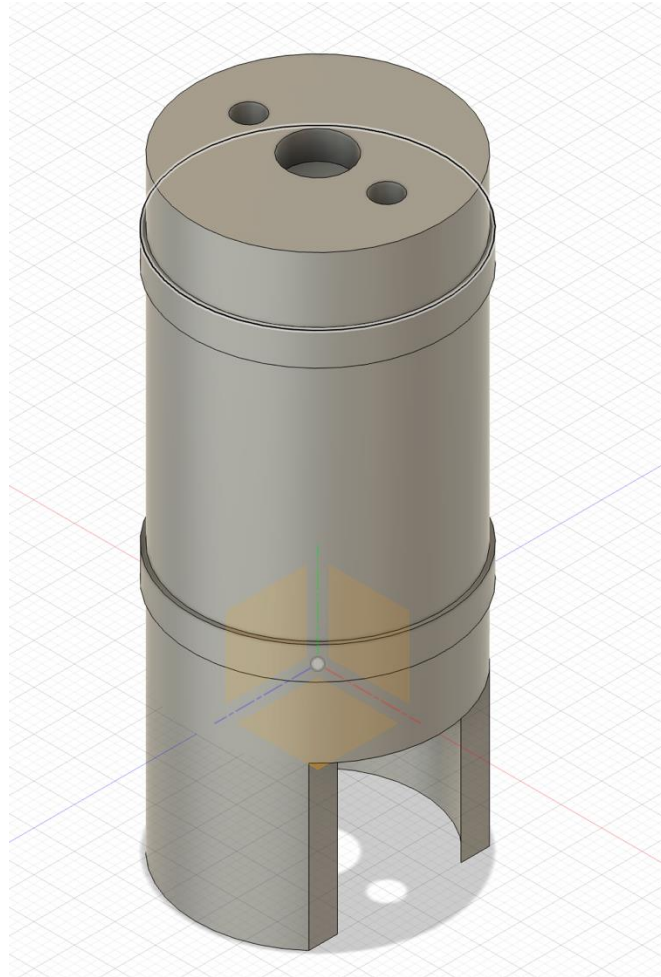
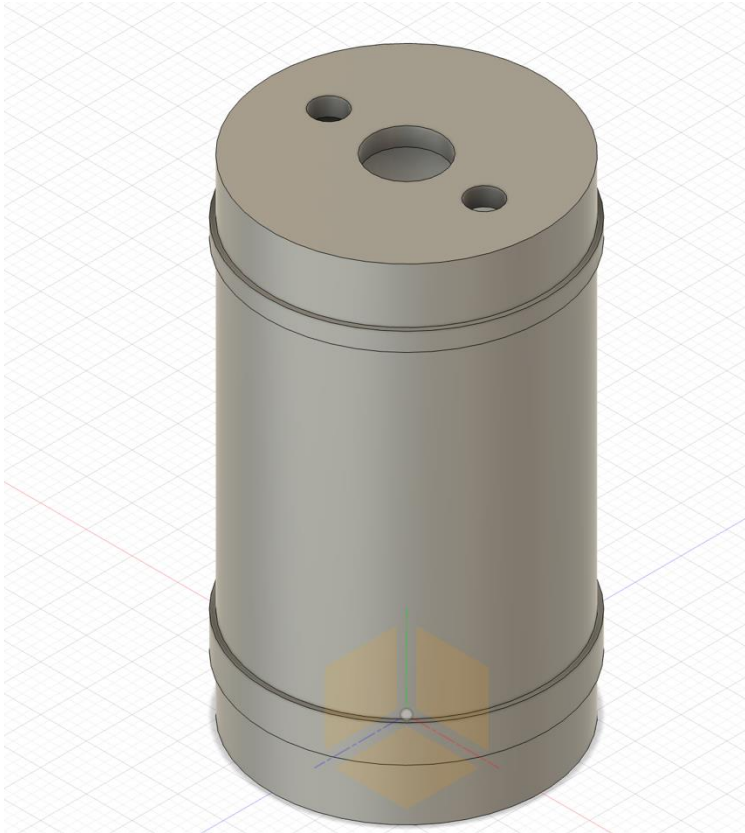
Other



Timeline



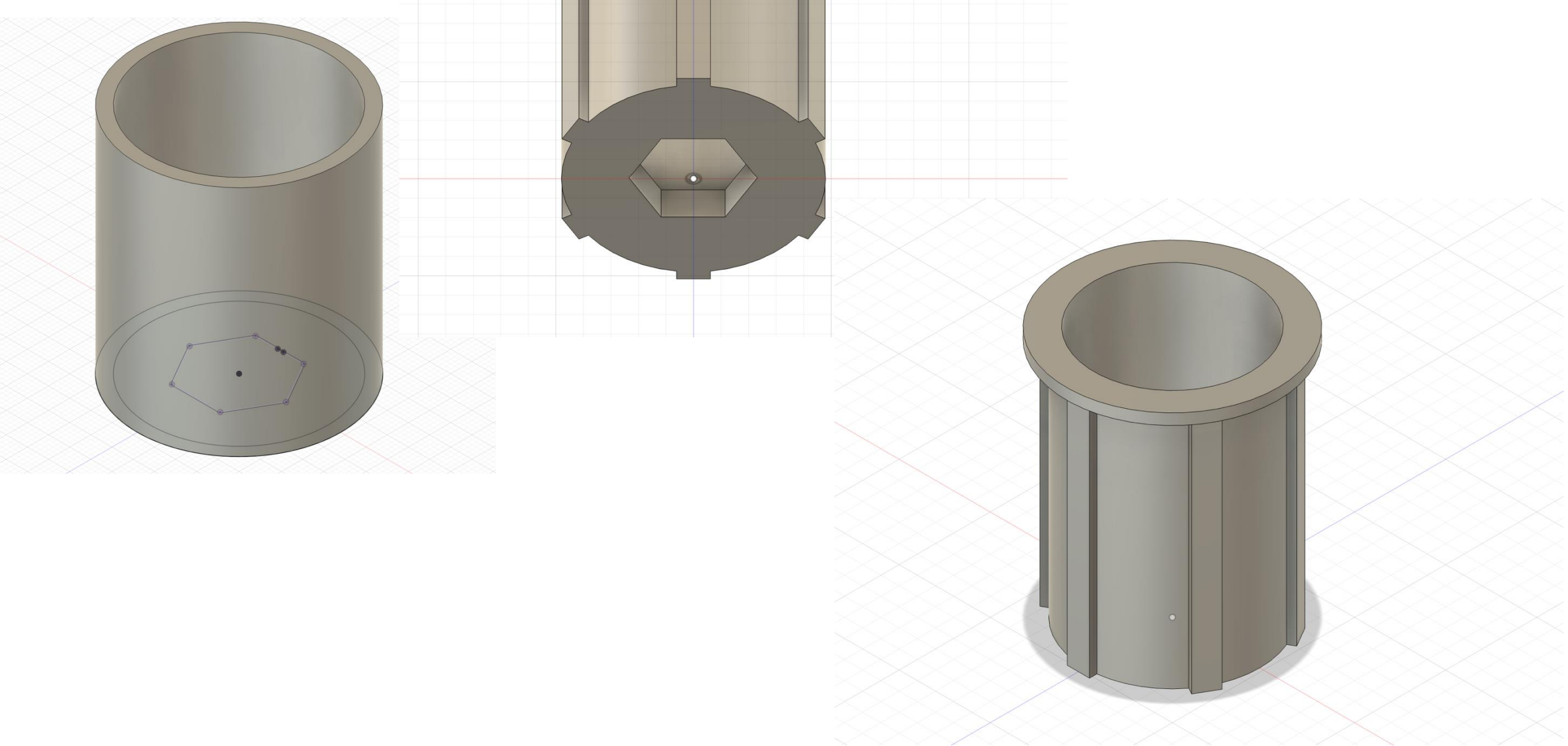
Mechanical



20 versions

 center hol...gcode.3mf	Sep 25, 2025 at 7:40 AM
 grooves sleeve.f3d	Sep 26, 2025 at 11:51 AM
 sleeve (25mm).stl	Sep 20, 2025 at 4:00 AM
 sleeve (26mm(.stl	Sep 20, 2025 at 4:00 AM
 sleeve (27mm).stl	Sep 20, 2025 at 4:01 AM
 sleeve 7.5...ter hole.stl	Sep 25, 2025 at 7:39 AM
 Sleeve 25.2 v2.stl	Sep 23, 2025 at 12:25 PM
 sleeve 25.4 v2.stl	Sep 23, 2025 at 12:26 PM
 sleeve 25.6 v2.stl	Sep 23, 2025 at 12:26 PM
 sleeve 25.8 vw.stl	Sep 23, 2025 at 12:27 PM
 sleeve 28mm.stl	Sep 20, 2025 at 4:01 AM
 sleeve ad....gcode.3mf	Oct 3, 2025 at 11:52 AM
 sleeve v3 25.2.stl	Sep 21, 2025 at 2:08 PM
 sleeve v3 25.4.stl	Sep 21, 2025 at 2:09 PM
 sleeve v3 25.6.stl	Sep 21, 2025 at 2:09 PM
 sleeve v3 25.8.stl	Sep 21, 2025 at 2:10 PM
 sleeve v4.gcode.3mf	Sep 25, 2025 at 4:19 AM
 sleeve v4.stl	Sep 25, 2025 at 4:18 AM
 sleeve v5.gcode.3mf	Sep 26, 2025 at 11:10 AM
 sleeve v5.stl	Sep 26, 2025 at 11:09 AM
 sleeve v6.stl	Sep 26, 2025 at 11:41 AM
 sleeve v10...motor.stl	Oct 3, 2025 at 11:51 AM
 sleeve v10...rooves.stl	Oct 3, 2025 at 11:59 AM
 sleeve wit...gcode.3mf	Sep 27, 2025 at 5:39 AM
 sleeve wit...oves v2.stl	Sep 27, 2025 at 5:38 AM
 sleeve wit...gcode.3mf	Sep 27, 2025 at 4:58 AM
 sleeve with grooves.stl	Sep 27, 2025 at 4:57 AM

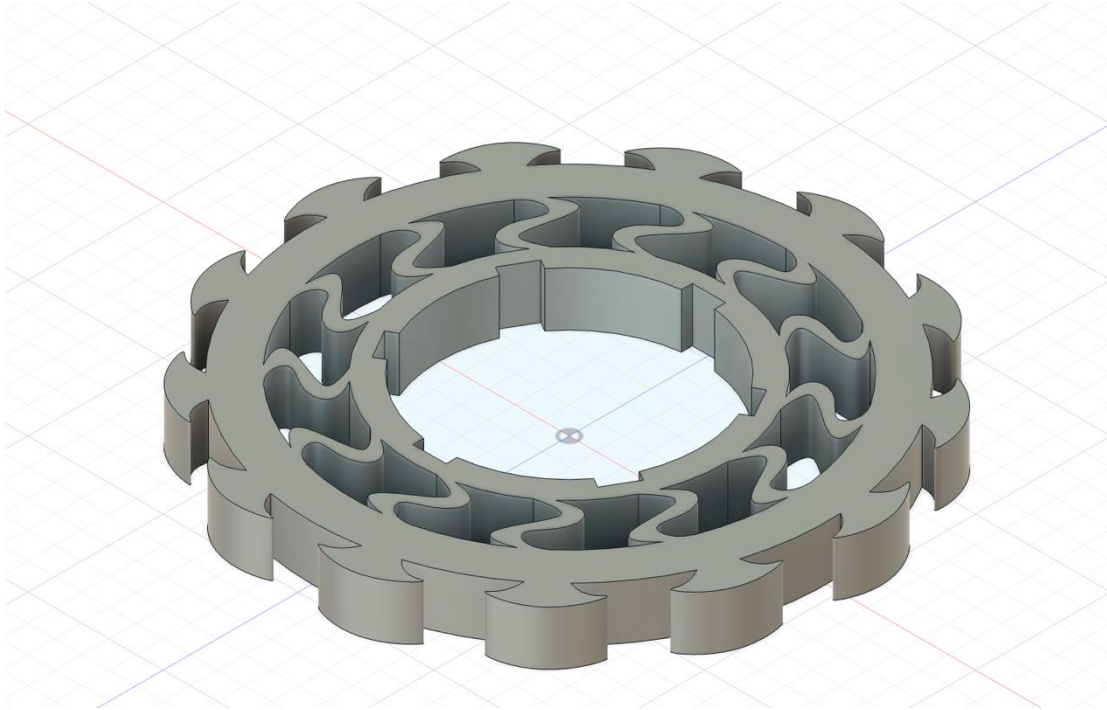




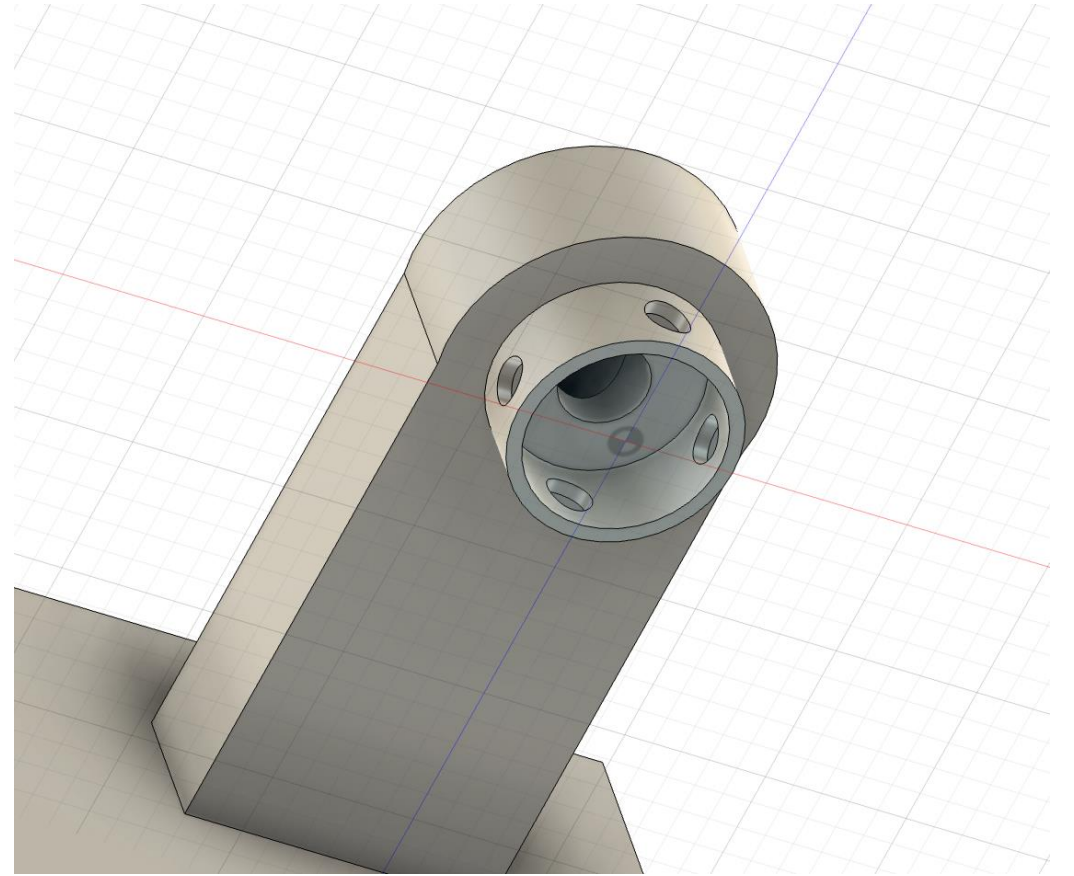
6 versions

steven.bang@stu.rosey.ch, laura.guo@stu.rosey.ch, haolin.liu@stu.rosey.ch, tianran.wang@stu.rosey.ch





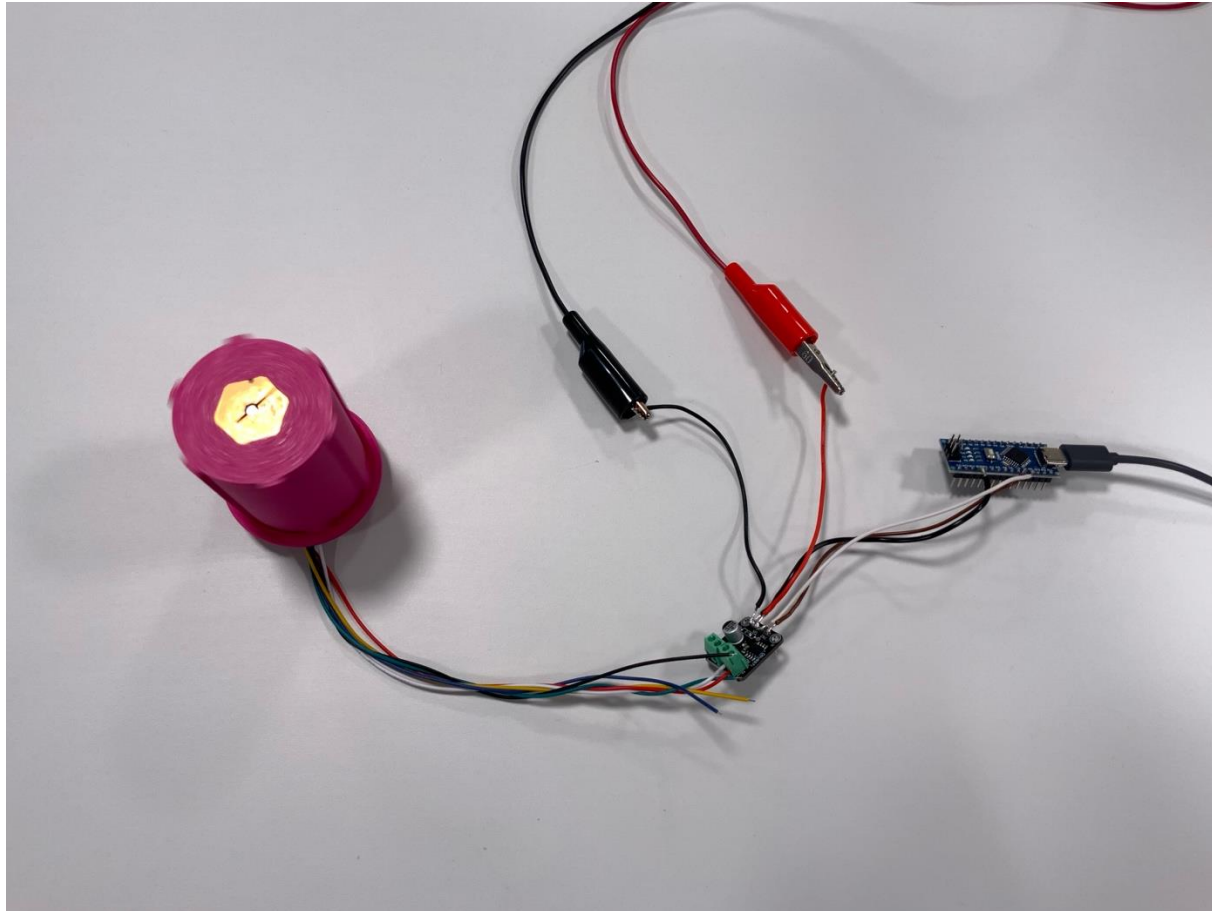
2 versions



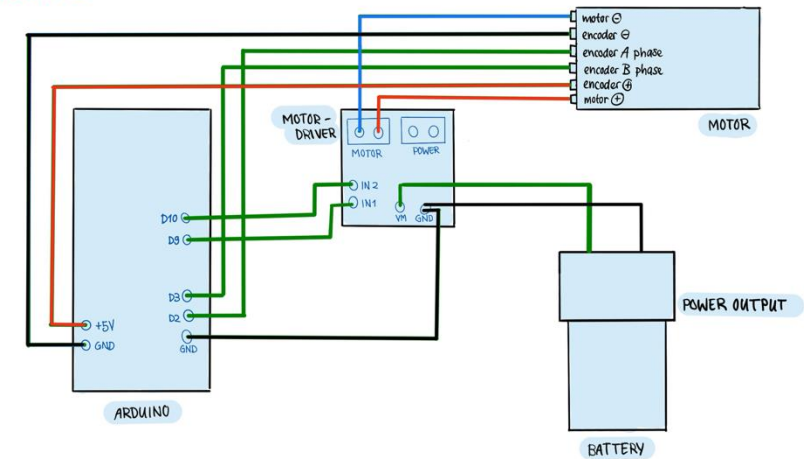
3 versions



Electrical



Electrical Schematics (draft 1)
option #2



Software

- Enterprise network limitations
 - Headless connection between laptop to Raspberry PI
 - Enterprise network connection in Raspberry PI
- Communication between Raspberry PI and Arduino
- Bluetooth connection with Controller
- Handling controller raw input
- Custom serial communication protocol
- Arduino pin alignment
- Raspberry PI missing software in Ubuntu 24.04.3 LTS



Software

In the foreseeable future

- Remote PI camera connection (limited by enterprise network)
- Controller input, buttons, and sensibility adjustment
- ROS2 refactoring
 - Peripheral host: actuator, sensor, interface, and utility nodes
 - Connection between each nodes: possibly limited by network
 - Possibly using `joy` for controller
 - Possibly using `raspicam\_node` for remote camera
 - Heard about `rviz2` for visualization, possibly need more sensors



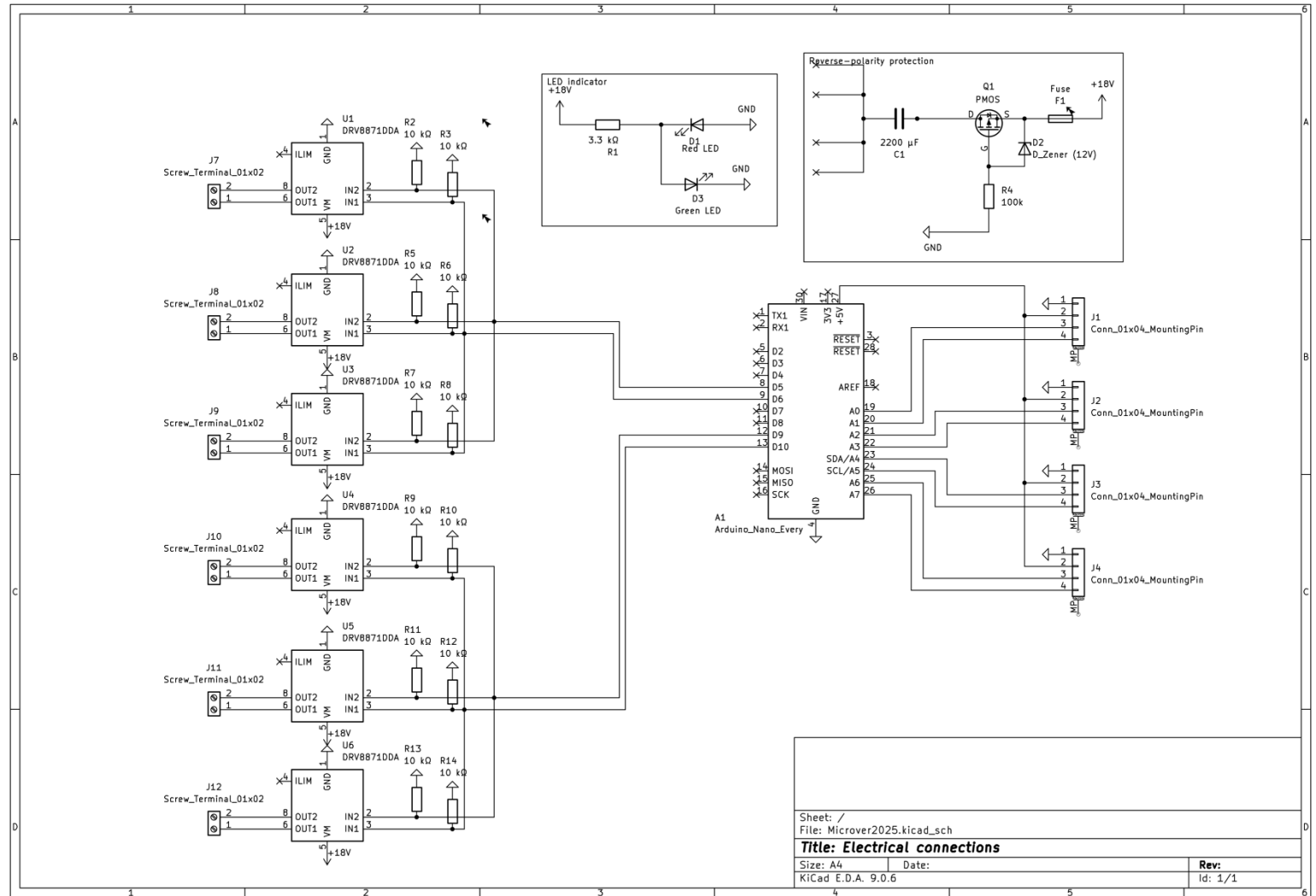
Management

Task Name	Progress	Assigned To	Created By	Created Date	Due date	Completed Date	Completed By
Mechanical - holding structure	Completed	BANG Steven	WANG Tianran	10/03/2025	10/10/2025	10/13/2025	BANG Steven
Electrical - Finish electronics connection with encoder pins	Not started	GUO Laura	WANG Tianran	10/10/2025	11/07/2025		
Mechanical - Finish sketch and fix size of sleeve	Completed	BANG Steven	WANG Tianran	09/19/2025	09/26/2025	09/26/2025	WANG Tianran
try out Pi camera	Completed	LIU Hao lin	WANG Tianran	09/26/2025		10/12/2025	WANG Tianran
Electrical - Started a sketch of the test bench design	Completed	GUO Laura	WANG Tianran	09/19/2025		09/19/2025	WANG Tianran
Electrical - Connect components according to schematic + start learning C++	Completed	GUO Laura	WANG Tianran	09/19/2025	09/26/2025	10/03/2025	WANG Tianran
Software - self directed	Completed	LIU Hao lin	WANG Tianran	09/19/2025	09/26/2025	09/19/2025	WANG Tianran
Software - connect Pi to controller via bluetooth + bidirectional communication	Completed	LIU Hao lin	WANG Tianran	09/23/2025	10/10/2025	10/10/2025	WANG Tianran
Mechanical - complete cad of hub	Completed	BANG Steven	WANG Tianran	09/26/2025	10/03/2025	10/04/2025	BANG Steven
For software engineering to test - Test bench Arduino with code	Completed	WANG Tianran	WANG Tianran	09/26/2025	09/26/2025	09/26/2025	WANG Tianran
Mechanical - Started a sketch of the test bench design + Learn parts	Completed	BANG Steven	WANG Tianran	09/19/2025	09/19/2025	09/20/2025	BANG Steven
Electrical - Finish code for test bench	Completed	GUO Laura	WANG Tianran	09/26/2025	10/03/2025	10/03/2025	WANG Tianran
Name and Logo	Completed	BANG Steven;GUO Laura;WANG Tianran;LIU Hao lin	CAREBIA Iulius	09/19/2025	09/26/2025	10/03/2025	WANG Tianran
Finish Testbench Presentation	Not started	BANG Steven;GUO Laura;WANG Tianran;LIU Hao lin	WANG Tianran	10/13/2025	10/31/2025		

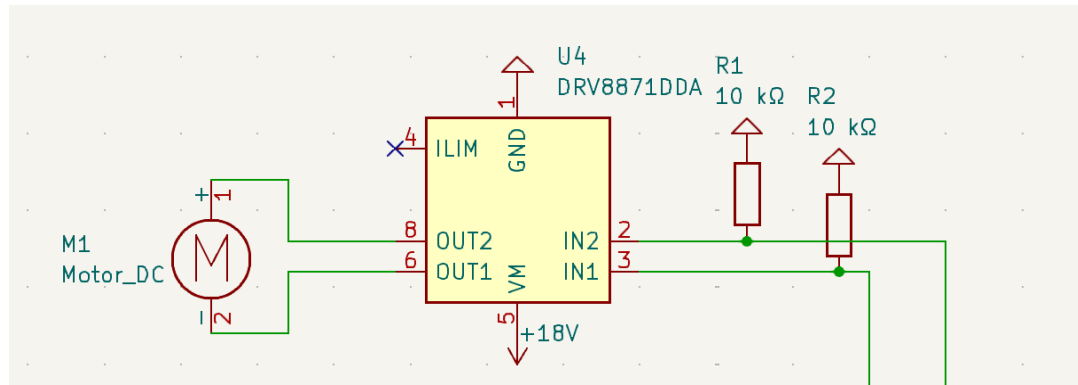
steven.bang@stu.rosey.ch, laura.guo@stu.rosey.ch, haolin.liu@stu.rosey.ch, tianran.wang@stu.rosey.ch



Schematics



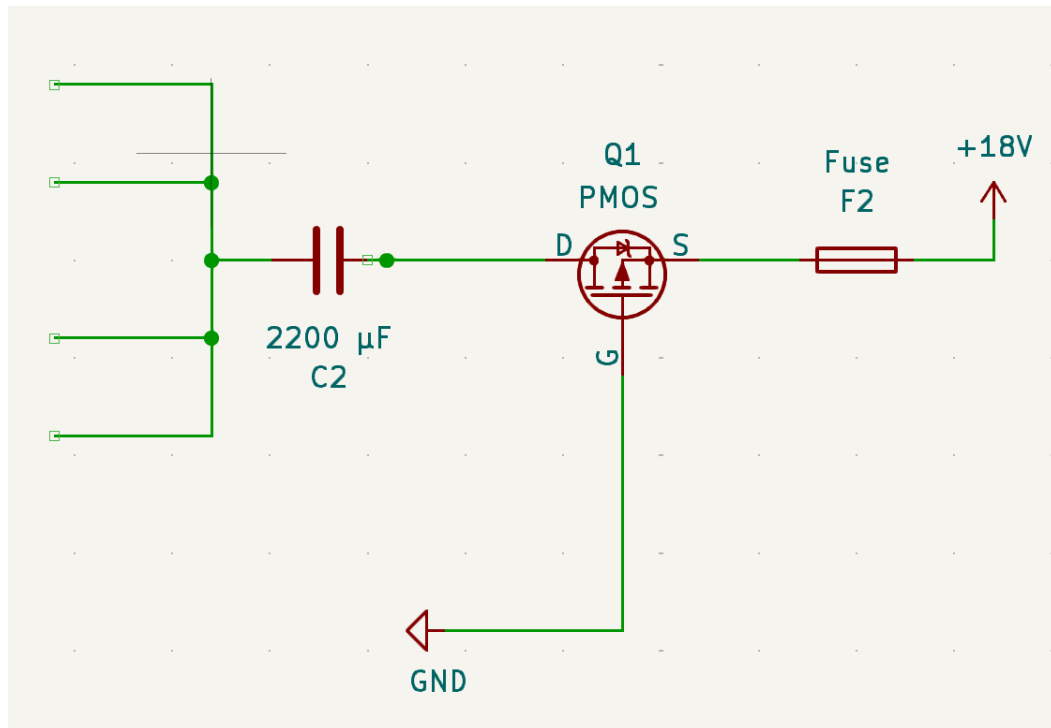
Pull-down resistor



$$I = \frac{5V}{10k\Omega} = 0.5mA$$

Reverse-polarity protection.

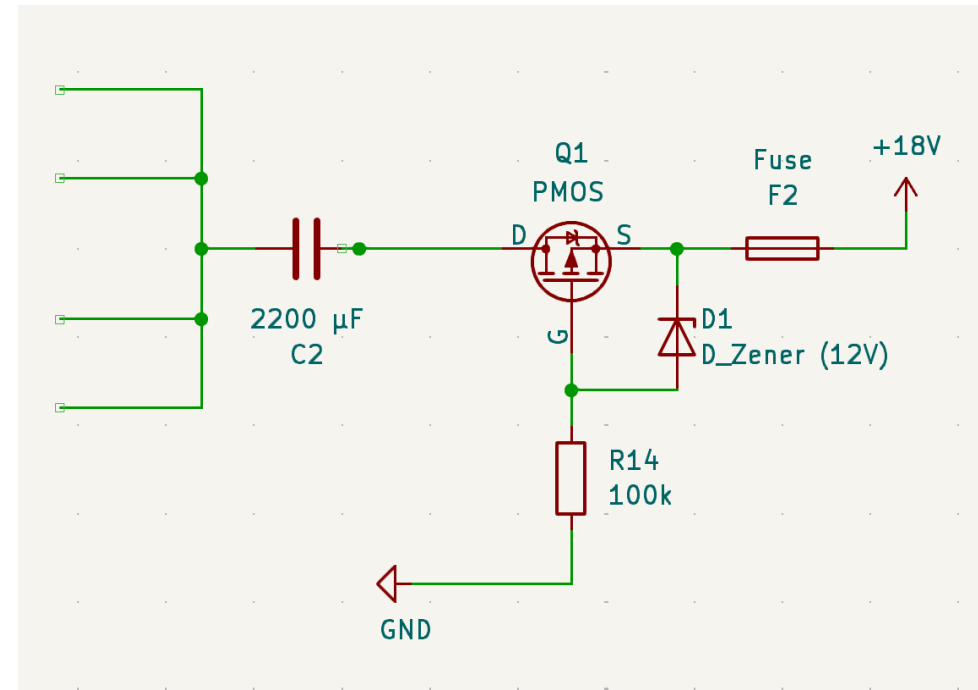
- MOSFET opens when voltage difference is < -4



$$V_{GS} = (V_G - V_S)$$

Zenor diode

- MOSFET's range of voltage (+-20V)
- Zenor diode's break down voltage -> reverse bias current-> limits voltage differences
- Increase in voltage -> increase in electric field at junction ->more electrons to break loose -> more current



$$P = V \times I = I^2 R = \frac{V^2}{R}$$

$$E = \frac{V}{d}$$

Where:

- E = electric field strength (in volts per meter, V/m)
- V = potential difference between the plates (in volts, V)
- d = distance between the plates (in meters, m)

LED indication

